

First year Aerospace Engineering

Mathematics - Calculus I

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Derivatives

Find the derivative of the following functions

1) $f(x) = \cos(\sin(3x))$

2) $f(x) = \tan(e^{2x^3})$

3) $f(x) = x^x$

Definition of the derivative

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Using the definition of the derivative, find the derivatives of the following functions:

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- 1) $\cos(x)$
- 2) e^{3x}
- 3) $\ln(x)$

Maxima and minima

Find the coordinates of the local maxima and local minima of the following functions

1) $e^{0.5x} \cos(2x), 0 < x < 4$

2) $x^3 - \frac{1}{3}x^4 + \sin(3x), 0 < x < 3$

3) $\frac{1}{4x} + \frac{1}{2}x^2 + 2x, -2 < x < 2$

4) $\frac{x}{x^2 + 2x - 3} + x$

Normal and tangent lines

Let $f(x) = x^3 - 3x^2$

$g(x)$ is the line parallel to $f(x)$ at $x = 1$

$h(x)$ is the line normal to $f(x)$ at $x = -1$

At what point (x, y) do the lines $g(x)$ and $h(x)$ meet?

Optimisation

Imagine a tube with water flowing inside. The tube has length π . The tube has varying diameter, so the speed of the water flowing inside also varies.

The speed of the water is given as a function of the length x as:

$$V = \sin(x) + \cos(2x) - \cos(x) + 1$$

- a) What is the maximum velocity the water achieves? At which point?
- b) What is the minimum? At which point?

You can install a turbine to generate electricity. The turbine works best at the point where the absolute acceleration of the water is largest ($F=m*a$ so largest force).

- c) At what point should you put the turbine? What is the acceleration of water at this point? What about the velocity?

Derivatives

Find the derivative of the following functions

$$1) \quad f(x) = \cos(\sin(3x)) = -3\sin(\sin(3x)) \cdot \cos(3x)$$

$$2) \quad f(x) = \tan(e^{2x^3}) = (6e^{2x^3} \cdot x^2) / (\cos^2(e^{2x^3}))$$

$$3) \quad f(x) = x^x = x^x(1 + \ln(x))$$

Maxima and minima

Find the coordinates of the local maxima and local minima of the following functions

1) $e^{0.5x} \cos(2x), 0 < x < 4$

Local maxima: (0.12, 1.03), (3.26, 4.96), Local minima: (1.69, -2.26)

2) $x^3 - \frac{1}{3}x^4 + \sin(3x), 0 < x < 3$

Local maxima: (0.62, 1.15), (2.44, 3.57), Local minima: (1.31, 0.56)

3) $\frac{1}{(4x)} + \frac{1}{2}x^2 + 2x, -2 < x < 2$

Local maxima: (-0.39, -1.35), Local minima: (-1.93, -2.13), (0.33, 1.47)

4) $\frac{x}{x^2 + 2x - 3} + x$

Local maxima: (-3.87, -4.78), (0.48, 0.21), Local minima: (-2.12, -1.25), (1.51, 2.17)

Normal and tangent lines

Let $f(x) = x^3 - 3x^2$

$g(x)$ is the line parallel to $f(x)$ at $x = 1$

$h(x)$ is the line normal to $f(x)$ at $x = -1$

At what point (x, y) do the lines $g(x)$ and $h(x)$ meet?

$(x, y) = 1.77, -4.31$

Optimisation

Imagine a tube with water flowing inside. The tube has length π .
The tube has varying diameter, so the speed of the water flowing inside also varies.

The speed of the water is given as a function of the length x as:

$$V = \sin(x) + \cos(2x) - \cos(x) + 1$$

- a) What is the maximum velocity the water achieves? At which point? $V = 3.1 \text{ m/s}$, $x = 2.94$
- b) What is the minimum? At which point? $V = 0.83 \text{ m/s}$, $x = 1.22$

You can install a turbine to generate electricity. The turbine works best at the point where the absolute acceleration of the water is largest ($F=m \cdot a$ so largest force).

- c) At what point should you put the turbine? What is the acceleration of water at this point?
What about the velocity? $x = 2.18$, $a = 2.125$, $V = 2.05 \text{ m/s}$